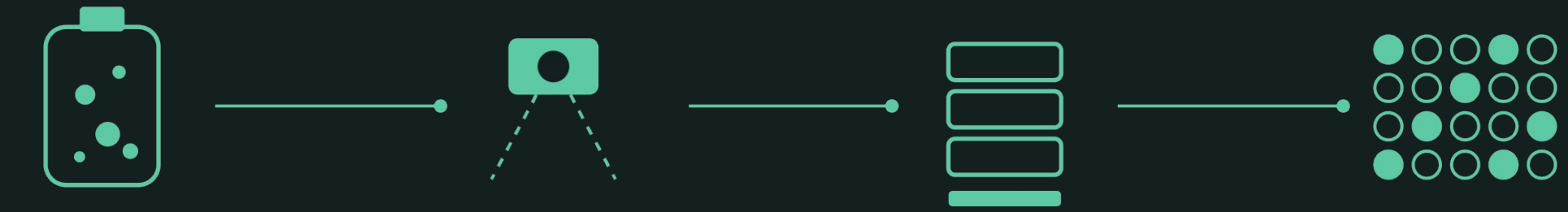
Hackathon pitch

Bags in, barcodes out

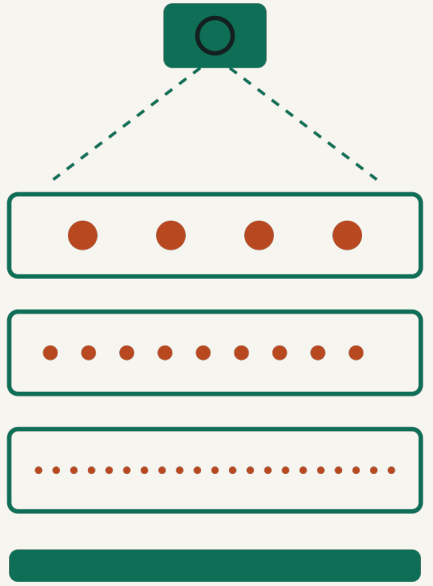
From digital twin to physical objects, every step under chain of custody · BioKEA Track A



Design intent

Sort a mixed sample with no manual sieving

- Camera measures the sample and picks the sieve sizes
- Robot arm builds the stack and handles every tray
- Loop of shake, image and check until each layer is pure
- Clean layers delivered to scientists in collection boxes



How it works

Six steps, each one checked

01

Observe

Camera and two sizers measure every body

02

Hypothesize

Pick sieve stack and water flow

03

Preflight

Digital twin must pass

04

Run

Arm places sieves, water rinses

05

Verify

Image each fraction, update margin

06

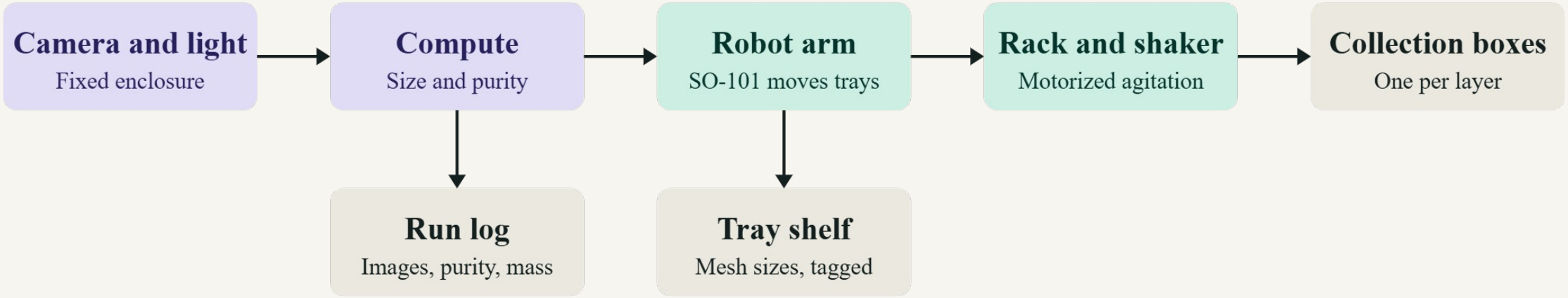
Hand off

Files for BugPicker, then wells

Every step is written to a hash-chained provenance log, per sample

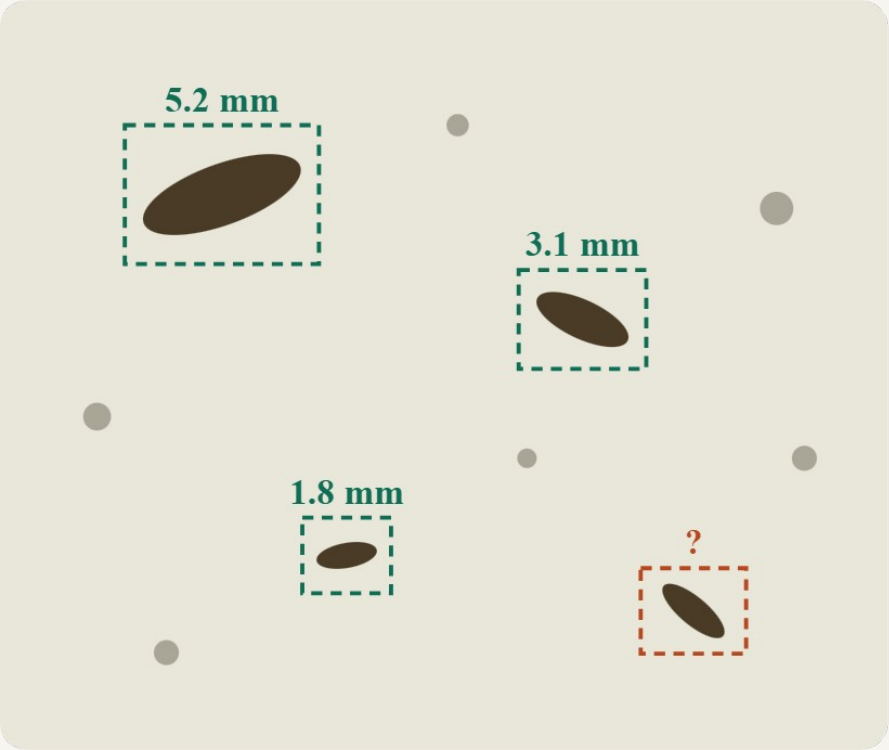
Design intent

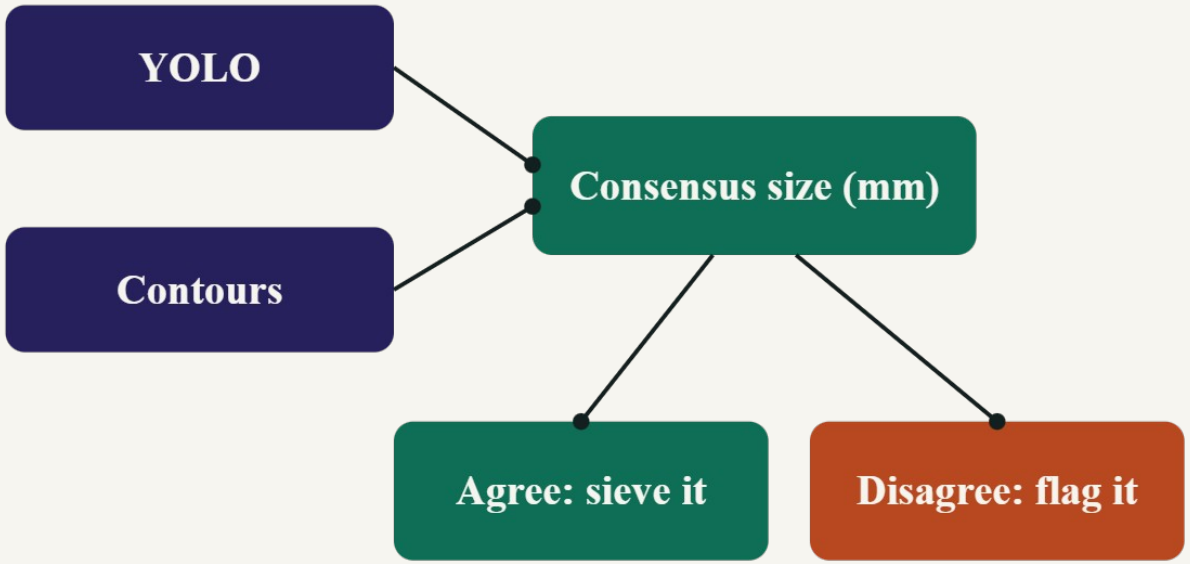
System architecture



Observe

Two sizers must agree

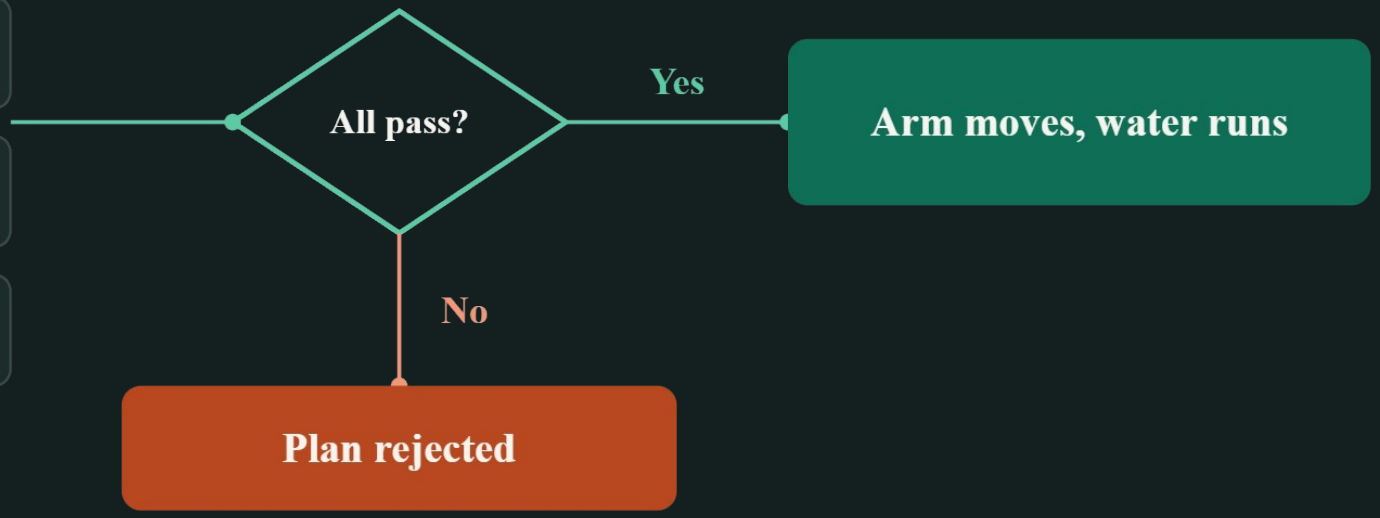




Preflight

Fail closed before hardware moves

- Sieve exists and is free
- Coarse to fine order
- Slots are available
- Water above reserve



Full lab stack

Digital to physical, one custody graph

- Filters: 32 hole types on one 50 mm plate (16 punched, 4 bought, 12 stainless)
- Plastic is the frame, metal is the aperture: 100 mm inserts and a 316L variable-aperture stage
- Vision at each handoff: YOLOE, OWLv2, the order classifier and a local VLM (qwen2.5vl:3b)
- Local models: 2,627 crops split by source photo feed small per-taxon preprocessors
- 91 printable PA12 toy insects, 1.0 to 29.9 mm, across 13 body plans
- 8 MIT labware remixes with Opentrons definitions, about $8.40 of filament
- Protocol Board: pin, run, remix or contest any protocol; every card points at exact bytes

Oct 24 to 25 · 489 commits · 31 PRs

Built in two days: the full stack

Physical

- 91 printed bug toys to train and test models
- Models also tested on toys and food
- Wood tower built by Wei, who also ran the printer
- Repurposed metal cut to fit robot trays
- SO-101 robot arm tested (PR #24)
- Two camera lanes: Andrew and Byron

Digital

- MCP server, CLI, three Vercel web apps
- Trained on an M1 Mac, 3 local preprocessors
- Python packages, datasets, deep research
- Custody graph: 1,175 leaves, 25 breaks
- Protocol Board, labware twin, demo reel
- 32 filters built for future runs

Repo record: 275 commits on Oct 24 and 214 on Oct 25, across PRs #1 to #31

Chain of custody

25 breaks recorded, not hidden

- Rule: every hash the graph references must resolve to stored bytes, otherwise it is a BREAK
- 1,175 hashed leaves in one graph pack, with 7 append-only MMR logs
- 8 results whose trained weights were never stored stay exploratory
- 16 bench photos and 1 bench source are pinned by hash but withheld (rights not established)
- Anyone can re-hash a file in the browser and contest a leaf as a GitHub issue
- The same pack feeds the web page, CLI, MCP server and npx

Design intent

Deciding when a tray is done

Purity per tray

Target objects divided by all objects in the tray image

Pass rule

Target purity set from calibration runs on labelled trays; not measured yet

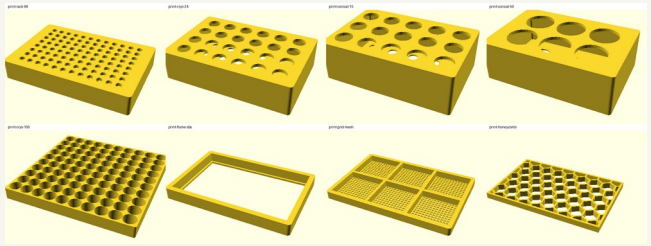
Stop and flag

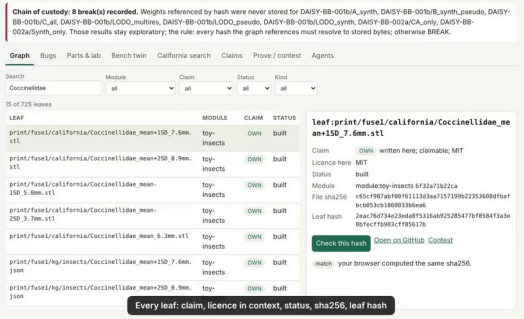
Cycle cap set at calibration; a failed gate is logged as a BREAK, then human review

Outcome

What we achieved

- Software end to end: sizing, sieve planning, twin preflight, custody graph, BugPicker files
- Real CA photos beat synthetic-only out of state: +0.696 Arizona, +0.521 Oregon (mask mAP50-95, exploratory)
- Order router outside California: macro-F1 0.70 vs 0.135 baseline, top-1 0.95 (n=231), weights stored
- Live: MVP graph and Protocol Board; physical cycle time not yet measured





Outcome

What we learned

- Hole shape matters: bodies lying flat pass round holes only near 0.6 to 0.7 of the diameter, so most pass end-on
- FDM PLA and PETG failed on the perforated trommel, so apertures moved to metal inserts in plastic frames
- The order model called bare floor Lepidoptera at 0.998: photo-vs-render checks come next
- Hymenoptera family model falls from 0.88 to 0.17 out of state, so it hands off rather than decides
- A kernel bug lost 8 runs of weights; weights are now hashed every epoch and checked fail-closed

Outcome

Future work

- Calibrate the variable-aperture stage and measure real mesh apertures
- Validate purity against hand-sorted or weighed trays; motorized shaker with the arm moving trays
- Species-level preprocessors, distilled to local models
- BioKEA novel-bug finder on Hugging Face
- Preprint on the custody method with a Heterodox Labs-style use case

Status

Built today, next on the bench

Built

- Camera sizing in mm, vision at each handoff
- Sieve planner, 32 filters, aperture stage
- Digital twin preflight and labware twin
- Custody graph, Protocol Board, MCP server
- 91 toys and 8 labware remixes, MIT

Next

- Drive the SO-101 arm and sieves
- Measure real apertures
- Close the loop on printed toys
- Collect real damage data

Ask: bench time on a real BugSorter run and labelled trays to calibrate purity, plus museum partners for species-level data